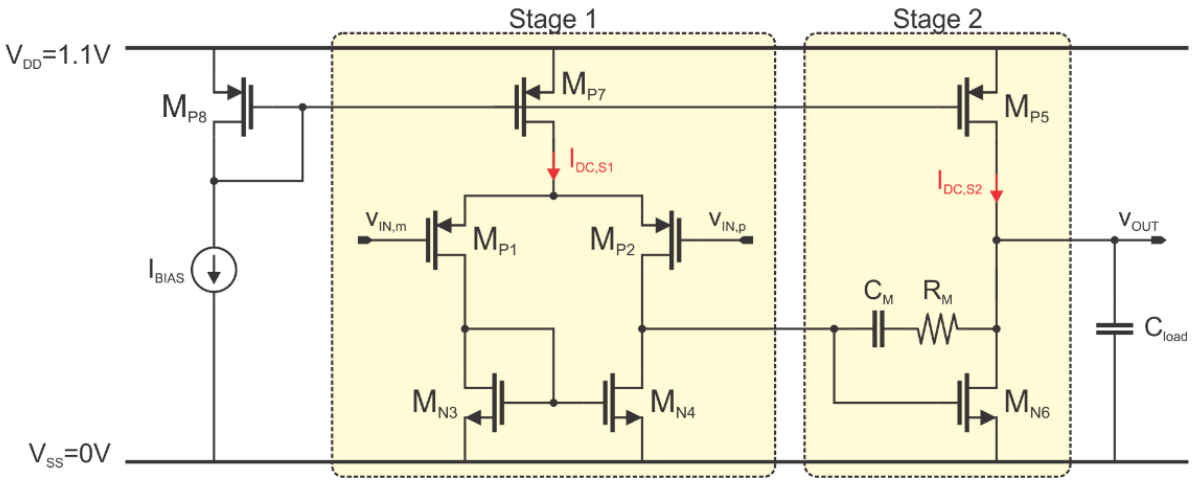
Analog Electronic Circuits – 2024-2025
Design Project Report

Group data

Group number	8
Name – Student 1	Chengping
Name – Student 2	

Goal:



Design of the 2stage OpAmp such that it passes the given specifications (insert your specs below):

$C_{load}[pF]$	10
DC gain [dB]	47
f_{GBW} [MHz]	50
Phase margin (PM) [deg]	> 70
Output swing [V]	> 0.7

Plan: Design of the 2-stage OpAmp on paper (10 points)

Calculate the V_{OV}, g_m, I_{DS} and L of each transistor (see exceptions under "*Remark*") and the required C_M and R_M in the OpAmp circuit. For that, insert all your calculations as well as your $\frac{g_m}{g_{ds}}$-, $\frac{g_m}{I_D}$-plots you used for your handcalculations below:

Hints:

- *First, plot $\frac{g_m}{g_{ds}}$ and $\frac{g_m}{I_D}$ across V_{OV} and gatelength L and do it again across V_{GS}, as presented in the 1st session. Think about whether to create the plots for a PMOS or for a NMOS device.*
- *Then, based on those plots, start your calculations.*
- *Furthermore, you can assume the following:*
  (1) the OpAmp is designed in triple-well-technology (i.e. $V_{SB} = 0$); (2) $V_{dsat} \approx V_{OV}$ and $V_{dsat} \geq 50mV$ across V_{OV}; (3) $C_M = \frac{1}{4} \cdot C_L$

Remarks:

- *For Mp5, Mp7 and Mp8 you are not required to calculate the g_m*

Plots used for the handcalculations:

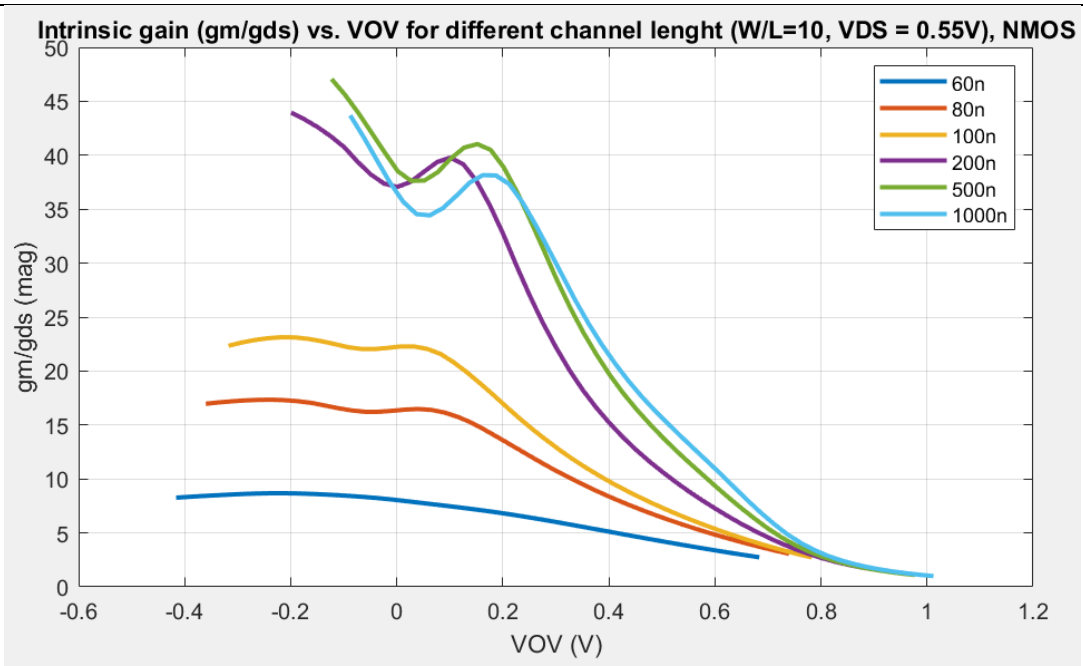


Figure 1: $\frac{g_m}{g_{ds}}$ vs V_{OV} across gatelength L

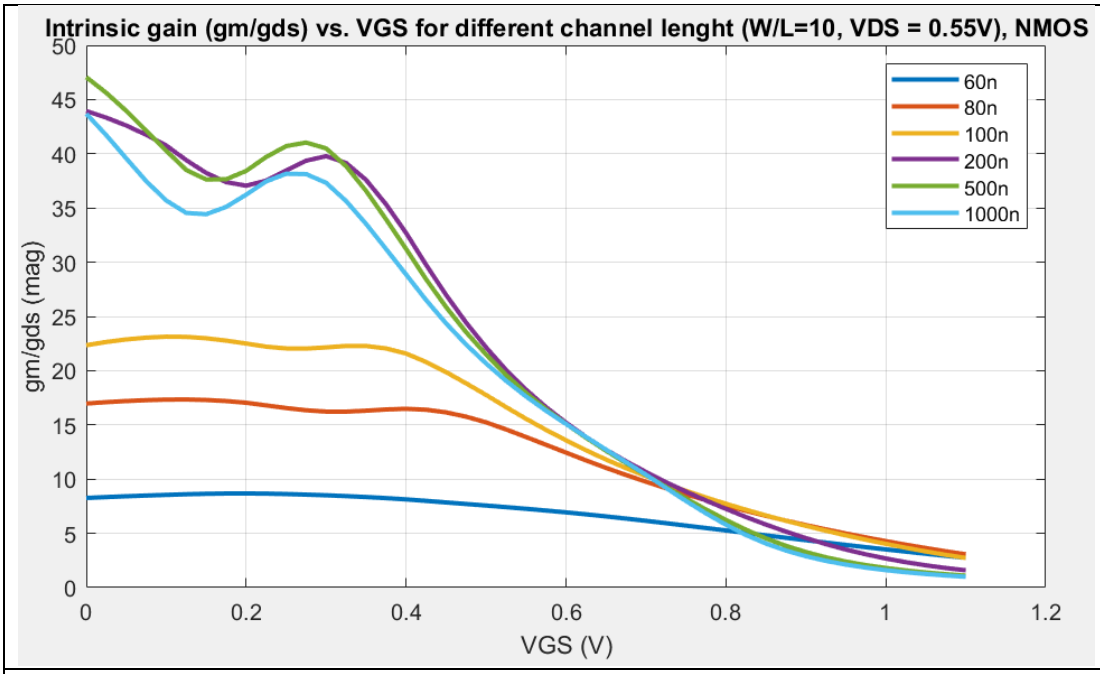


Figure 2: $\frac{g_m}{g_{ds}}$ vs V_{GS} across gatelength L

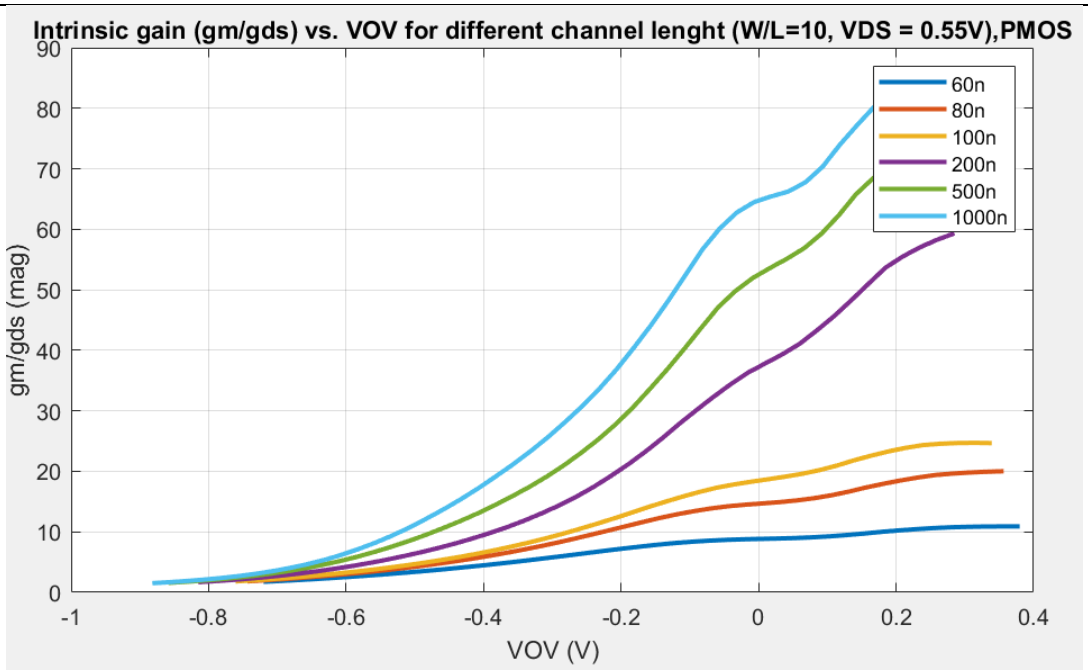


Figure 3: $\frac{g_m}{g_{ds}}$ vs V_{OV} across gatelength L

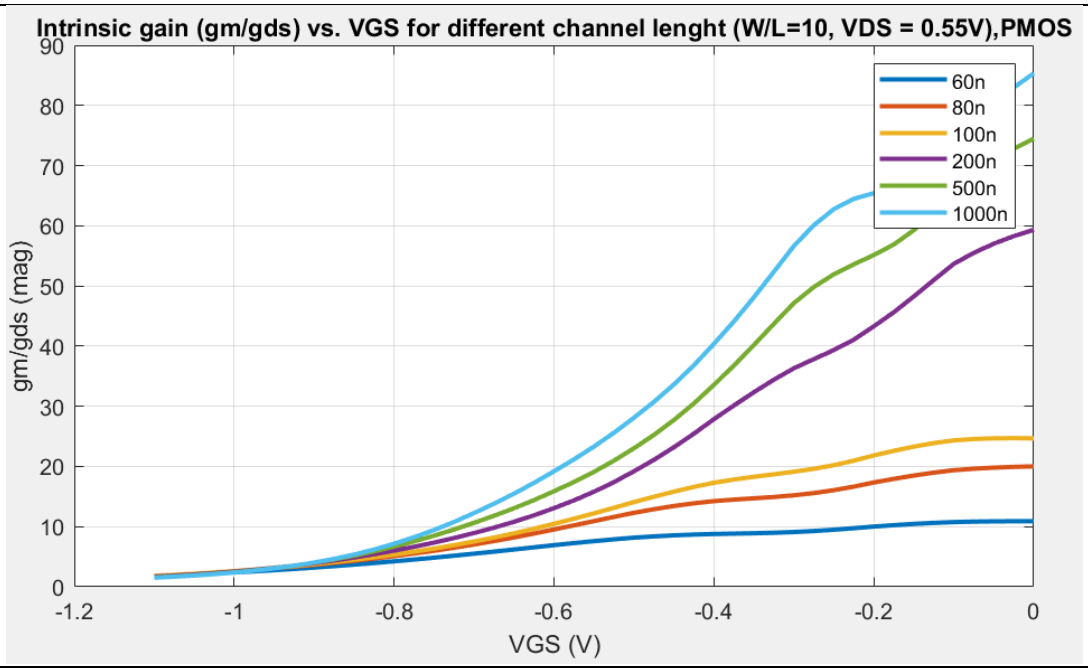


Figure 4: $\frac{g_m}{g_{ds}}$ vs V_{GS} across gatelength L

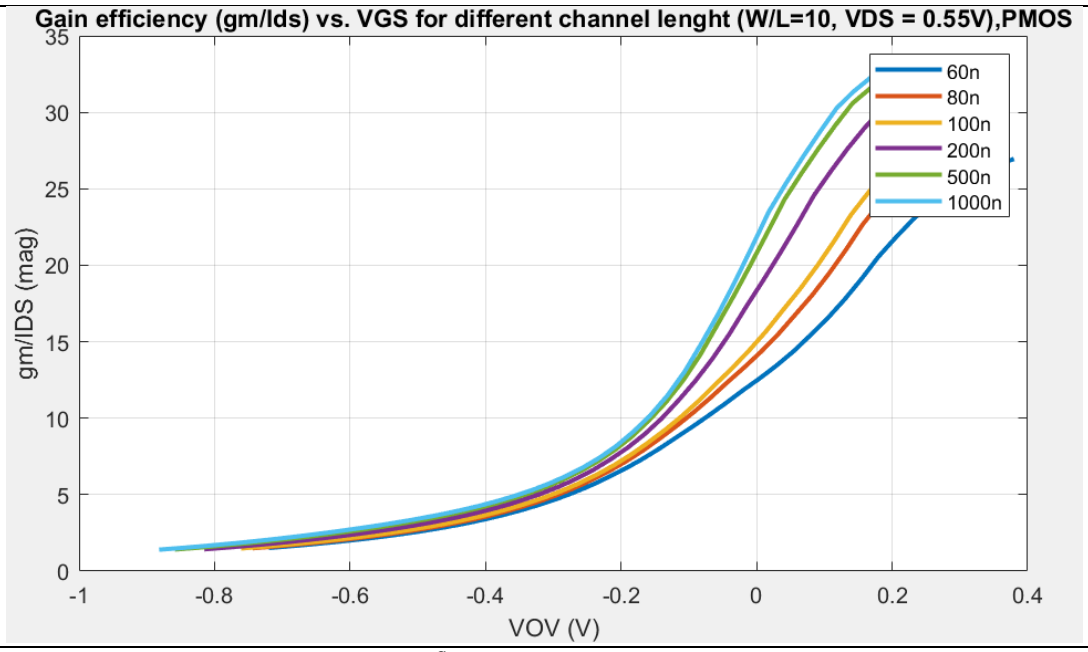


Figure 5: $\frac{g_m}{I_D}$ vs V_{OV} across gatelength L

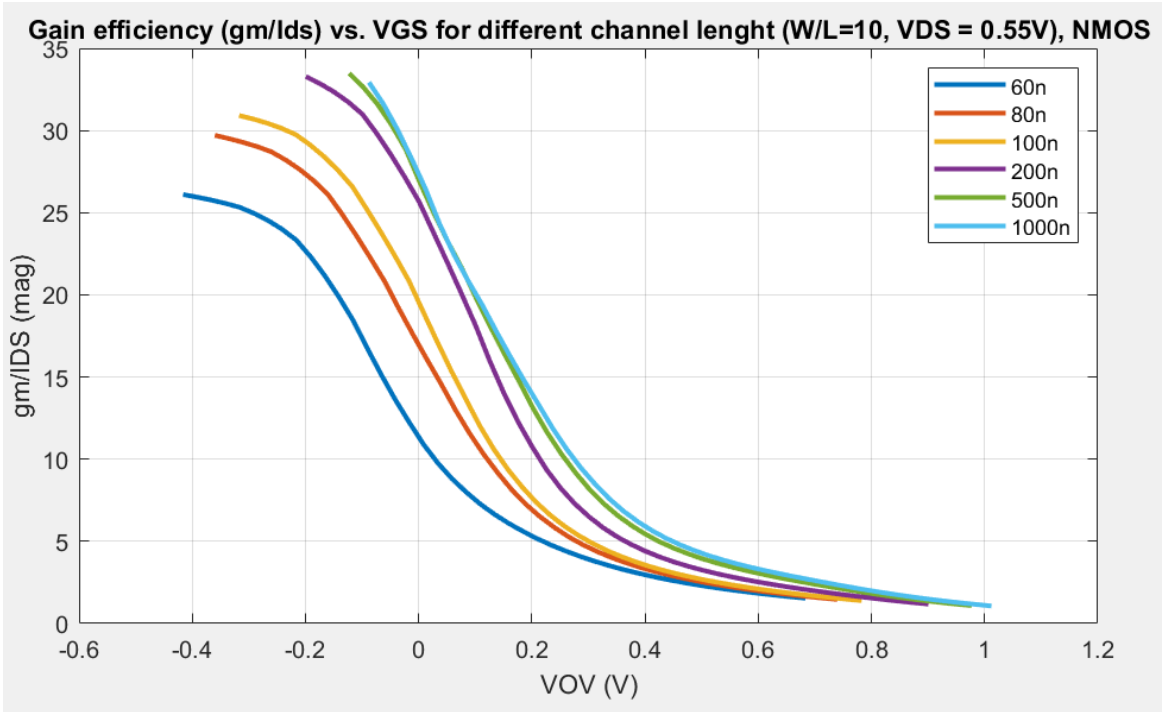


Figure 6:

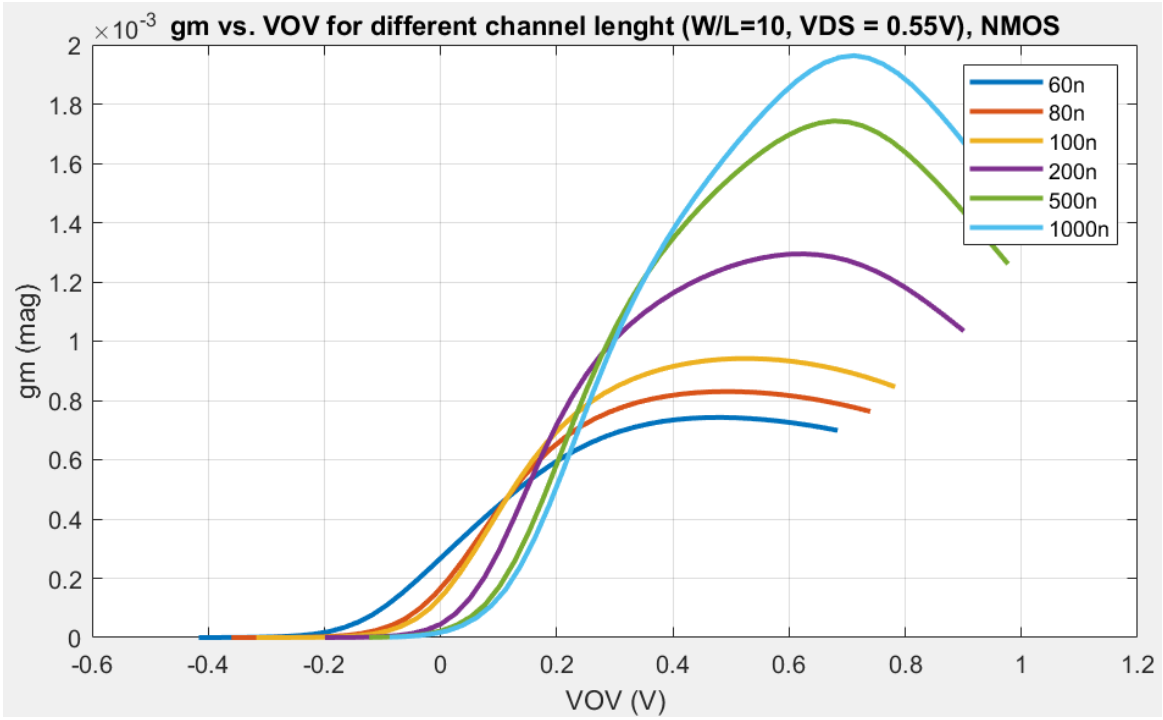


Figure 7:

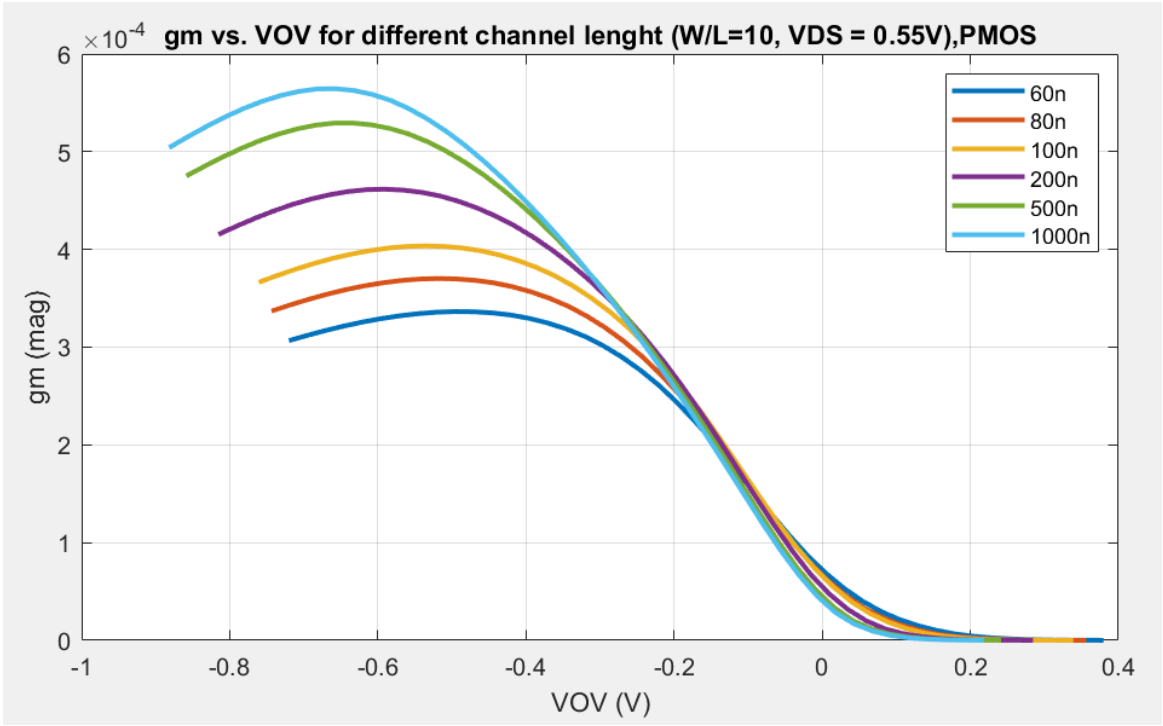


Figure 8:

Handcalculations:

New Version: i have two version of my Hand calculations. My old version have problem, but new version is not finished.

I set the Mp8, Mp7, Mp5 in the weak inversion because they need to apply the stable current. I set the Mp1, Mp2 and Mn6 in the strong inversion because of the function of amplifier, but the length of Mp1 and Mp2 is smaller than the Mn6 because the noise is easy to amplifier in the first stage. And Mn3, Mn4 are better in the weak inversion because they need to apply the stable current, but the real case have some constraints.

Old version: dB=20×log10 (Amplitude), Amplitude=10^(dB/20)

$f_{GBW} = 50\ MHz$, $CL = 10\ pF$ $PM = 80$ degree

$227.87 \times 2\pi$

$A_{V1} =$

DC gain $A_V = A_{V1} \times A_{V2} = 47\ dB = 10^{A(47/20)} = 223.87$ mag.

CM is upon, target cap. in M_6

$$A_{V1} = \frac{g_{m2} + g_{mb2} + g_{m3}}{g_{ds4} + g_{ds2} + g_{gs4} + g_{gs2} + g_{gs6} + g_{gd6}}$$

does $r_o = \frac{1}{g_{ds}}$ same as Pmos or Nmos

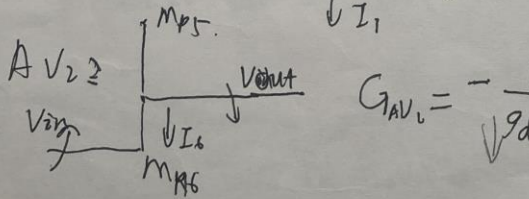
$$A_{V2} = \frac{g_{m6} + g_{mb6}}{g_{ds6} + g_{ds5} + g_{d1}}$$

why $A_{V1} =$?

$$V_{out} = I \cdot R_{out} = \frac{I}{g_{out}} = \frac{I_1 + I_2}{g_{ds2} + g_{ds4}} = \frac{\left|\frac{V_{in}}{2}\right| \cdot g_{m1} + \left|\frac{V_{in}}{2}\right| g_{m2}}{g_{ds2} + g_{ds4}} = \frac{V_{in} \cdot g_{m1,2}}{g_{ds2} + g_{ds4}}$$

$G = \frac{V_{out}}{V_{in}} = \frac{-g_{m1,2}}{g_{ds2} + g_{ds4}}$, we can set $g_{ds2} = g_{ds4}$, simple to select the size of M_{p2}, M_{n4},

A_{V1} actually node: I_2 ↑, V_{out} ↓, I_1 ↓ so it's negtive.

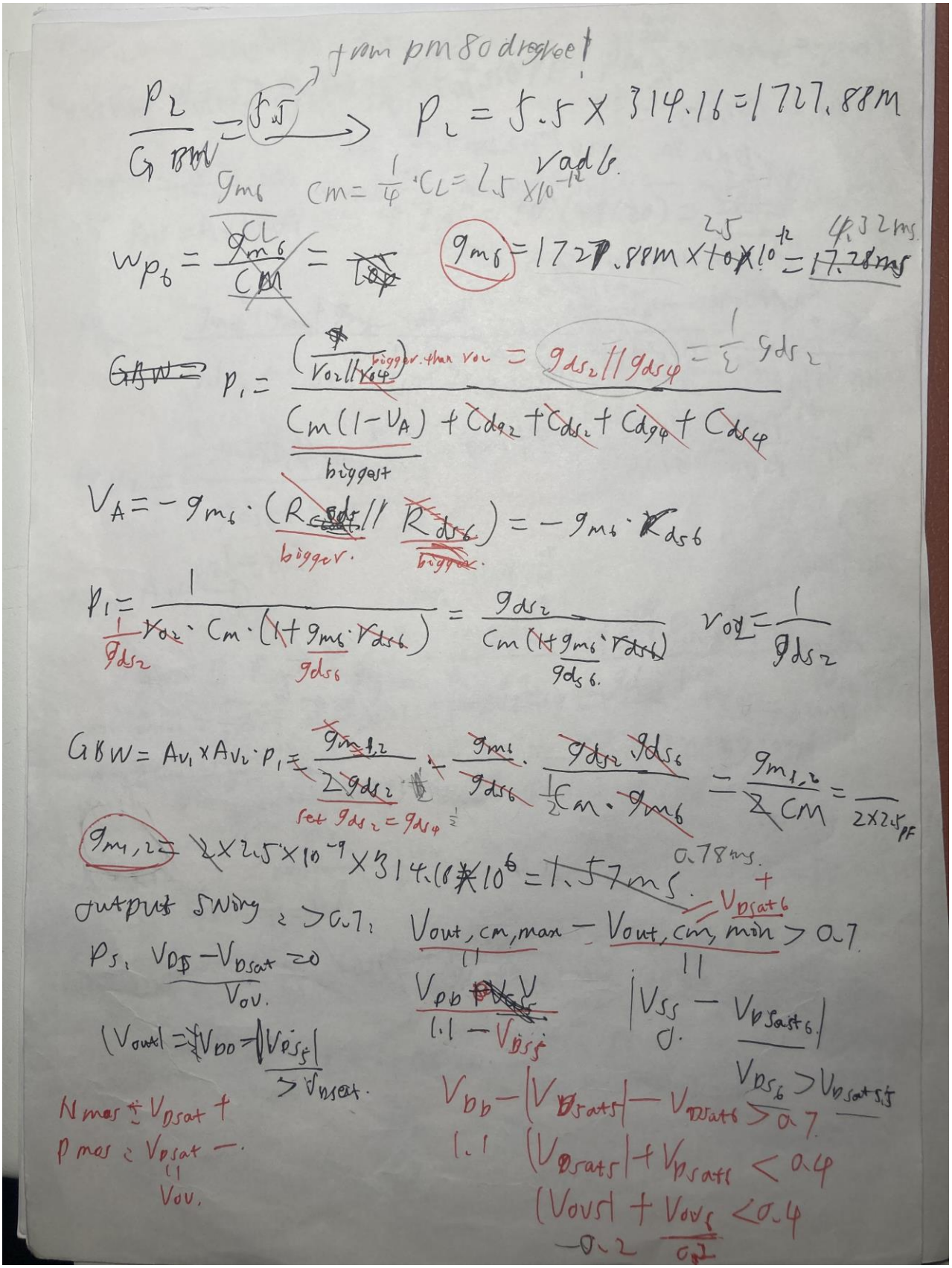


$G_{AV_2} = -\frac{g_{m6}}{g_{ds5} + g_{ds6}}$ so assume $g_{ds5} = 0$, simple to cal.

$GBW = \frac{g_m}{g_{mC}}$ $2\pi \cdot f_{GBW} = 2 \cdot pi \times 50M = 314.16M$ rad/s

$GBW = A_{V1} \times A_{V2} \cdot P_1$ $P_1 = 1.4 \times 223.3 \times 10^3$ rad/s

gain bandwidth


from pm 80 degree!
$\frac{P_2}{G_{BW}} = 5\sqrt{}$ → $P_2 = 5.5 \times 314.16 = 1727.88M$
vadb.
g_{m6} $C_M = \frac{1}{4} \cdot C_L = 2.5 \times 10^{-12}$
$w_{p6} = \frac{g_{m6}}{C_M}$ $g_{m6} = 1727.88M \times 2.5 \times 10^{-12} = 4.32ms$
$P_1 = \frac{(\frac{1}{r_{o2} \| r_{o4}})}{C_M(1-V_A) + C_{dg2} + C_{ds2} + C_{dg4} + C_{ds4}}$ bigger than r_{o2} $= g_{ds2} \| g_{ds4} = \frac{1}{2} g_{ds2}$
biggest
$V_A = -g_{m6} \cdot (R_{ds4} \| R_{ds6}) = -g_{m6} \cdot r_{ds6}$
bigger. bigger.
$P_1 = \frac{1}{\frac{1}{g_{ds2}} \cdot C_M \cdot (1 + g_{m6} \cdot r_{ds6})} = \frac{g_{ds2}}{C_M(1 + \frac{g_{m6}}{g_{ds6}} \cdot r_{ds6})}$ $r_{o2} = \frac{1}{g_{ds2}}$
g_{ds2} g_{ds6} g_{ds6}.
$GBW = A_{V1} \times A_{V2} \cdot P_1 = \frac{g_{m1,2}}{2 g_{ds2}} \cdot \frac{g_{m6}}{g_{ds6}} \cdot \frac{g_{ds2} g_{ds6}}{\frac{1}{2} C_M \cdot g_{m6}} = \frac{g_{m1,2}}{2 C_M} = \frac{}{2 \times 2.5 pF}$
set $g_{ds2} = g_{ds4}$ $\frac{1}{2}$
$g_{m1,2} = 2 \times 2.5 \times 10^{-9} \times 314.16 \times 10^6 = 1.57ms$ 0.78ms.
output swing > 0.7: $V_{out,cm,max} - V_{out,cm,min} > 0.7$
$V_{DB} - V_{DSat}$ $V_{DD} - V_{DSat6} + V_{DSat6}$ $|V_{SS} - V_{DSat6}|$
P_5: $V_{DS} - V_{DSat} > 0$ V_{ov} 1.1 − V_{DSat5} 0.
V_{ov}
$|V_{out}| = |V_{DD} - |V_{DS5}|| > V_{DSat}$
$V_{DS6} > V_{DSat5}$
Nmos $\geq V_{DSat}$ +
Pmos $\geq V_{DSat}$ −
V_{ov}
$V_{DD} - |V_{DSat5}| - V_{DSat6} > 0.7$
1.1
$|V_{DSat5}| + V_{DSat1} < 0.4$
$|V_{ov5}| + V_{ov1} < 0.4$
−0.2 0.2

$g_{m6} = 4.32ms$ $g_{m1,2} = 0.78ms$

we hope $|V_{ov5}| + V_{ov6} < 0.4$, so set $V_{ov5} = -0.2V$

also want to keep V_{ov5} in strong inversion, $V_{ov6} = 0.18V$.

$N V_{ov6} = 0.18V \xrightarrow{\text{length}} 80 \xrightarrow{\frac{g_m}{g_{ds}}} 14 \xrightarrow{g_m} 4.32ms \xrightarrow{\frac{g_m}{I_{ds}}} 7.6 \xrightarrow{I_{ds}} 0.568mA \rightarrow 568\mu A$

$I_{d8} = 565mA$

$P_{ov5} = -0.2V \xrightarrow{\text{length}} 1000 \xrightarrow{\frac{g_m}{g_{ds}}} 7.26 \xrightarrow{\frac{g_m}{I_{ds}}} 6.6 \xrightarrow{I_{ds}} 0.568mA \xrightarrow{g_m} 3.75ms$

$P_{ov5} = P_{ov7} = P_{ov8} = -0.2V \xrightarrow{\text{length}} 1000nm \xrightarrow{g_m} 3.75ms$

$A_{v1} = 5.94$ mag

$V_{gs6} = V_{ds3} = V_{ds4} = V_{gs3} = V_{gs4} = 0.55V$

$N_{N4} = N_{V_{gs4}} = 0.55V \xrightarrow{\text{length}} 500nm \xrightarrow{\frac{g_m}{I_{ds}}} 5mag \xrightarrow{g_m} 1.18 \xrightarrow{V_{ov}} 0.4V$

N_{N3}

$N_{P2} = \frac{g_m}{g_{ds2}} = 5.94mag \xrightarrow{100} 100 \xrightarrow{V_{ov}} -0.43 \xrightarrow{0.78} I_{ds} \rightarrow 0.236mA \rightarrow 236\mu A$

$200 \rightarrow 200 \xrightarrow{V_{ov}} -0.51 \xrightarrow{\frac{g_m}{I_{ds}}} 2.9 \xrightarrow{0.71} I_{ds} \rightarrow 0.27mA \rightarrow 270\mu A$

$I_{d3} = 2I_{ds} = 472\mu A$

N_{P1}

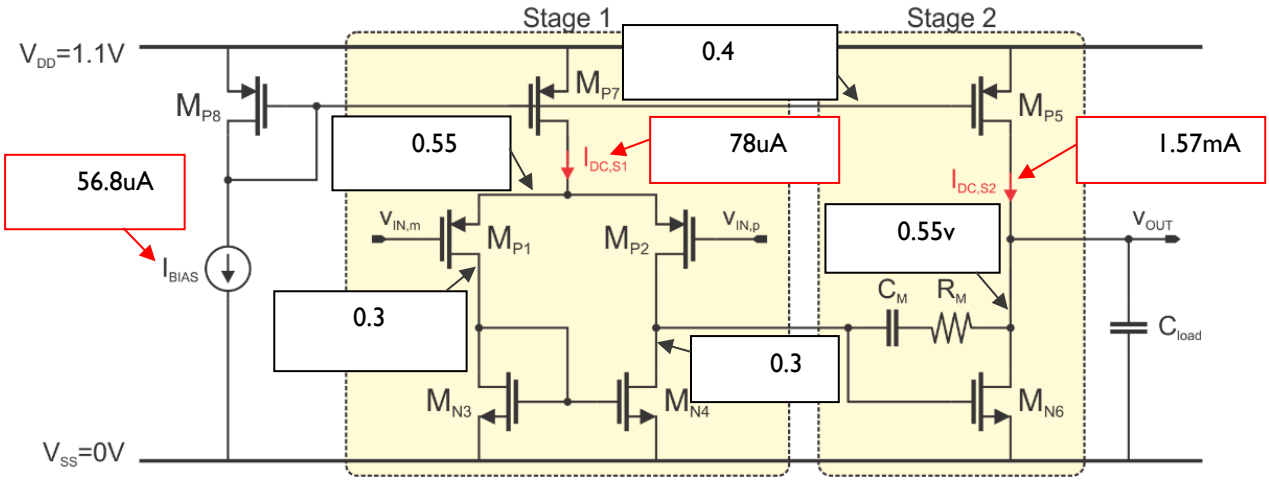
Place the obtained values in the table below. Only the values in here need to be calculated / chosen in the handcalculations. Each value should be chosen with a reason which you have to provide above.

Device	L [nm]	I_{ds} [µA]	V_{ov} [V]	g_m [S]
M_{p1}	100	236	-0.43	0.78 *10^-3
M_{p2}	100	236	-0.43	0.78 *10^-3
M_{n3}	500	236	0.4	1.18 *10^-3
M_{n4}	500	236	0.4	1.18 *10^-3
M_{p5}	1000	568	-0.2	
M_{n6}	80	568	0.18	4.32*10^-3
M_{p7}	1000	568	-0.2	
M_{p8}	1000	568	-0.2	

Device	Units	Value
R_M	Ω	-0.22
I_{BIAS}	A	56.8uA

Place all the calculated voltages (in the black box) and all currents (in the red box) on the circuit depicted below and calculate also (include formula):

$V_{cm,in,min}$	0+Vgs3-Vth1=0.55+0.128=0.678
$V_{cm,in,max}$	1.1-(Vgs7-Vth7)=1.1+0.2=1.3
$V_{out,min}$	0+Vov6=0.18v
$V_{out,max}$	Vdd+Vov5=1.1-0.2=0.9v
P_{diss}	$VDD*(ISS + Ibias)$


Stage 1
Stage 2
V_{DD}=1.1V
V_{SS}=0V
M_{P8}
M_{P7}
M_{P5}
M_{P1}
M_{P2}
M_{N3}
M_{N4}
M_{N6}
I_{BIAS}
$I_{DC,S1}$
$I_{DC,S2}$
$V_{IN,m}$
$V_{IN,p}$
V_{OUT}
C_M
R_M
C_{load}
0.4
0.55
78uA
1.57mA
56.8uA
0.55v
0.3
0.3

Design: Implementation of the OpAmp in MATLAB and LTspice (10 points):

Based on your handcalculations, create your OpAmp design in MATLAB.

Remarks:

- *Please make sure that all widths W are below 1mm*
- *Don't write your values calculated in the previous step as constants in MATLAB but write the full equation in your code.*
- *Values chosen as design choices should match between this table and the one from the handcalculations!*
- *Choose Vincm to be in the center of your input common-mode range. Do this in a 2-step process:*
  - *Use the center of the input common-mode range as calculated in the handcalculations for Vincm.*
  - Have MATLAB calculate the correct Vincm value using the calculated 'vdsat' fields of the transistors in MATLAB.

Fill out both tables depicted below. All values are to be determined in MATLAB.

Device sizes and bias point parameters according to MATLAB

Device	W [µm]	L [nm]	I_{ds} [µA]	V_{ov} [V]	g_m [S]	g_{ds} [S]	g_m/g_{ds} [-]	$V_{ds,sat}$ [V]	V_{ds} [V]
M_{p1}	4.47	100	326.45	-429.99m	863.94u	1.1m	785.36m	-356.76m	-200m
M_{p2}	4.47	100	326.45	-429.99m	863.94u	1.1m	785.36m	-356.76m	-200m
M_{n3}	6.33	500	326.45	418.56m	1.66m	131.44u	12.6	295.82m	400m
M_{n4}	6.33	500	326.45	418.56m	1.66m	131.44u	12.6	295.82m	400m
M_{p5}	1.02	1	2.58	-200m	20.82m	3.88m	5.36	-175.4m	-200m
M_{n6}	25.3	80	2.58	179.99m	19.6m	1.38m	14.2	150.57m	550m
M_{p7}	215.48	1	652.89	-200m	5.52m	159.32u	34.65	-175.24m	-500m
M_{p8}	18.46	1	56.8	-200m	478.92u	12.68u	37.77	-175.21m	-550m

Device	Units	Value
C_M	pF	2.50pF
R_M	Ω	-173.70 Ohm

I_{BIAS}	A	0.06mA
$V_{in,cm}$	V	Vin,cm,min [mV] -181 Vin,cm,max [mV] 129

Based on the parameters filled in the table above, design your OpAmp in LTspice. The filled-in MATLAB template and final LTspice schematic should be included in your Canvas submission (read the assignment instructions on Canvas to see the full description of the files to include).

Experiment (10 points):

In this section you will simulate the following:

A. Frequency Response in MATLAB and LTspice
B. Noise contribution in LTspice
C. Linearity in LTspice

Note on how to present graphs in general:

- *When you make a graph, put labels on every axis to make clear what you are showing. Also, show the units!*
- *When you plot multiple curves on one graph, add a legend.*

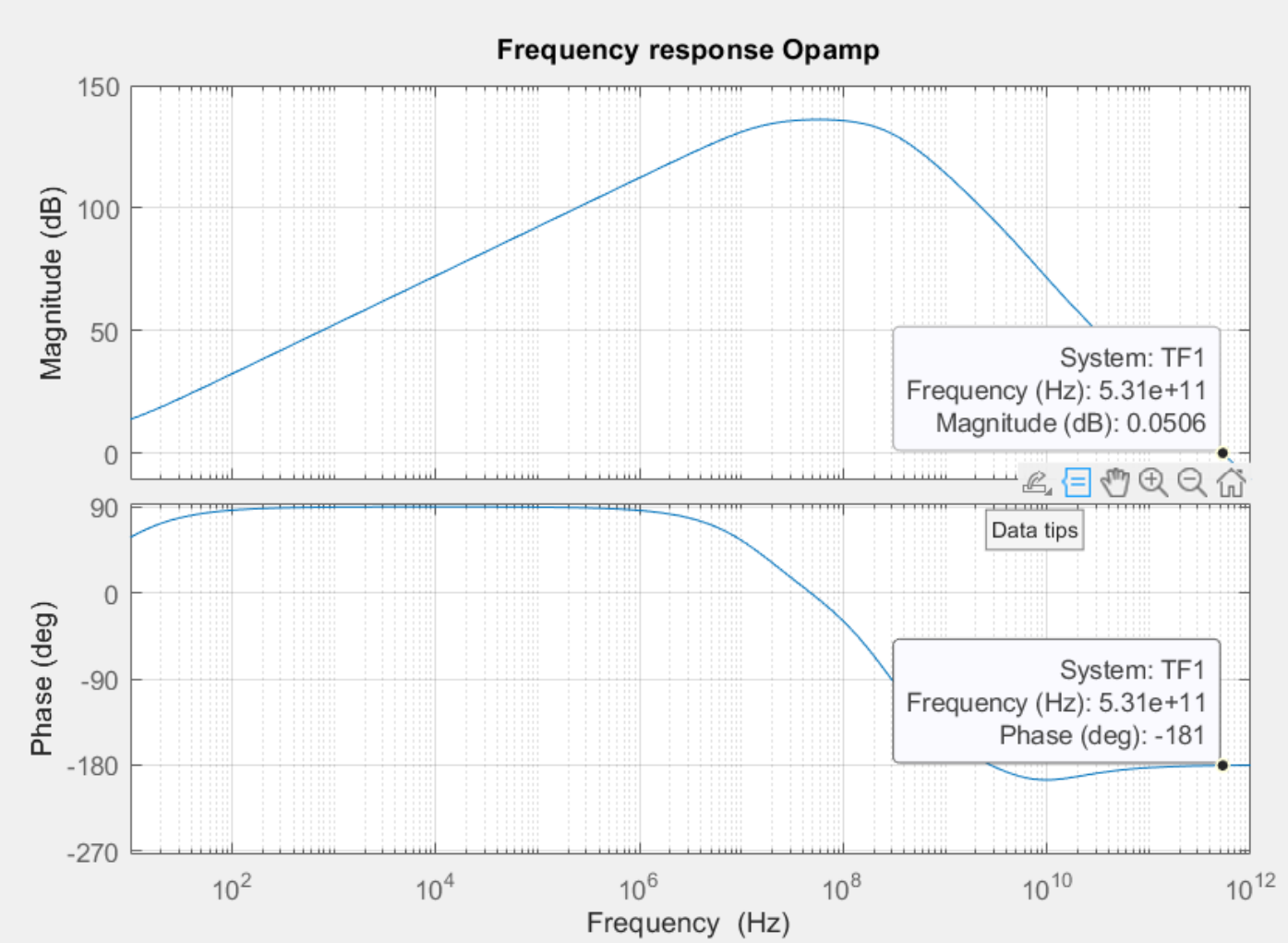
A. Frequency Response in MATLAB and LTspice

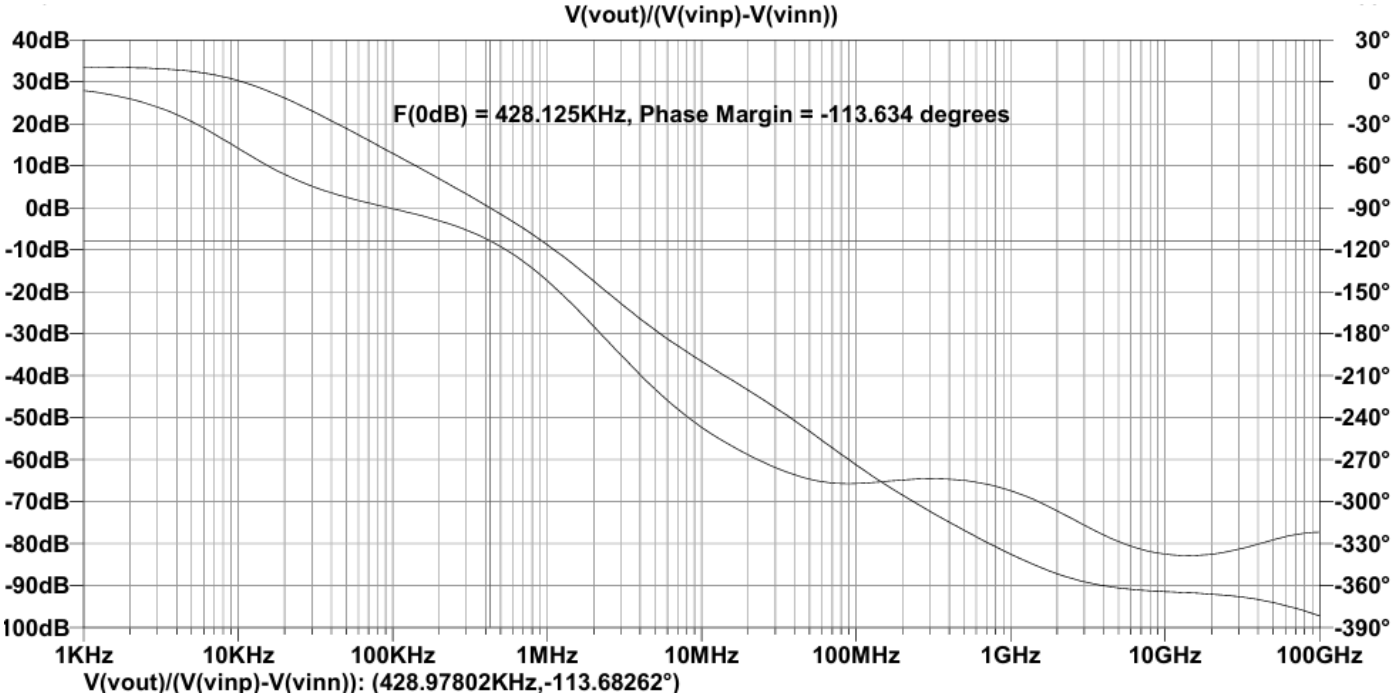
Note on how to present graphs in this sub-section:

- ***For magnitude plots use dB (linear scale) vs. Hz (logarithmic scale)***
- ***for phase plots use ° (linear scale) vs. Hz (logarithmic scale).***

Simulate A_v (small voltage gain) with C_M and R_M in MATLAB and LTspice. Then, paste both A_v-curves in separate plots below such that the difference between MATLAB and LTspice can be clearly seen. In addition to that, indicate the resulting phase margin (PM) in both plots.

Plots:


Frequency response Opamp
Magnitude (dB)
150
100
50
0
System: TF1
Frequency (Hz): 5.31e+11
Magnitude (dB): 0.0506
Data tips
Phase (deg)
90
0
-90
-180
-270
System: TF1
Frequency (Hz): 5.31e+11
Phase (deg): -181
10^2
10^4
10^6
10^8
10^{10}
10^{12}
Frequency (Hz)



Simulator	PM (deg)
MATLAB	1
LTspice	67

Explain, analyse and interpret the results above. Furthermore, if you observe significant differences between MATLAB and LTspice, explain why.

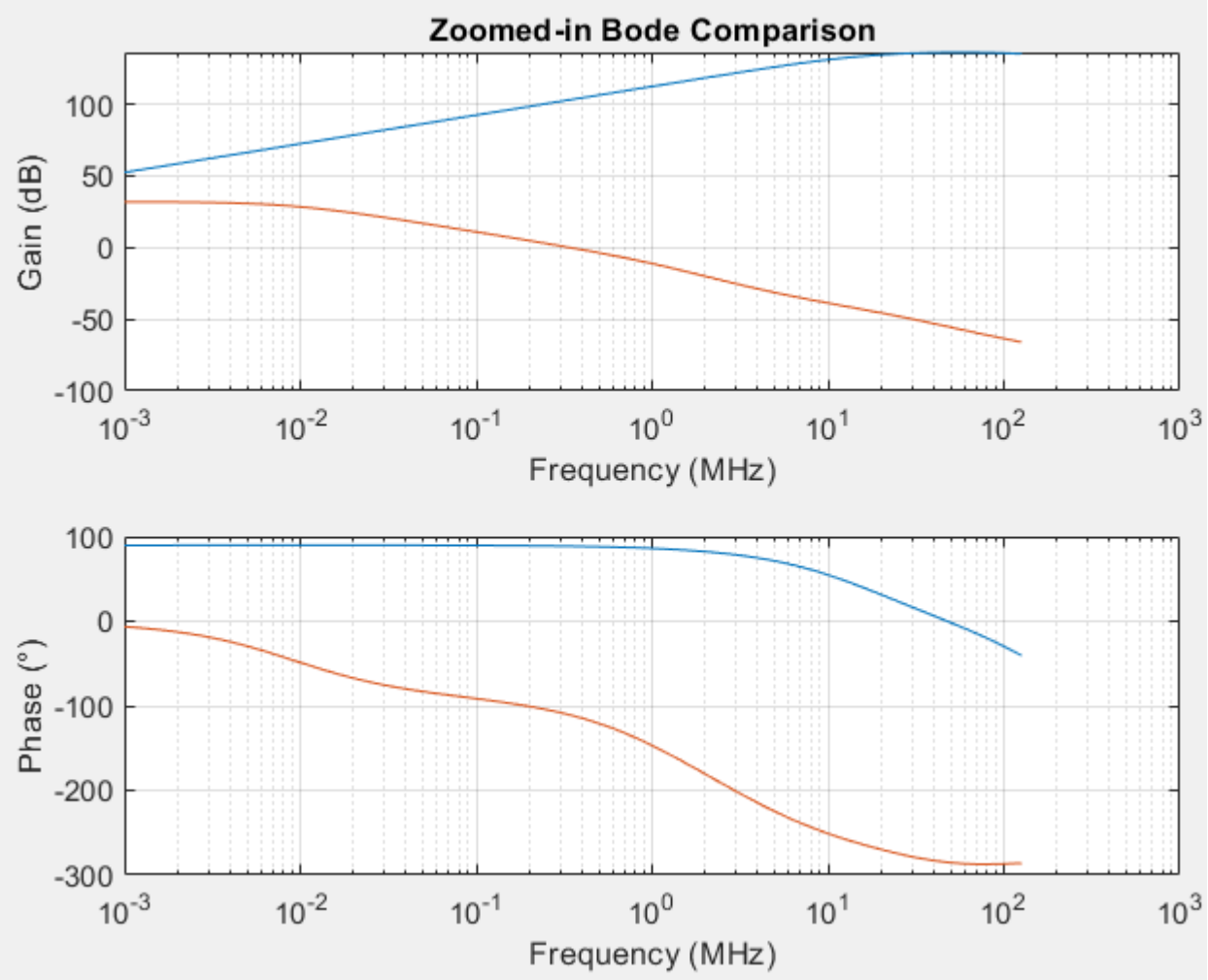
I get the matlab gragh is really strange. The result is not matched with the our design requirements. The phase should be above 70 degree, but the result is 1 degree in the phase. I think I set the wrong zero to cancel the wrong poles, so the magnitude is going up and the resistance is negative. I should use the zero to cancel the pole in the node 2 in the matlab.

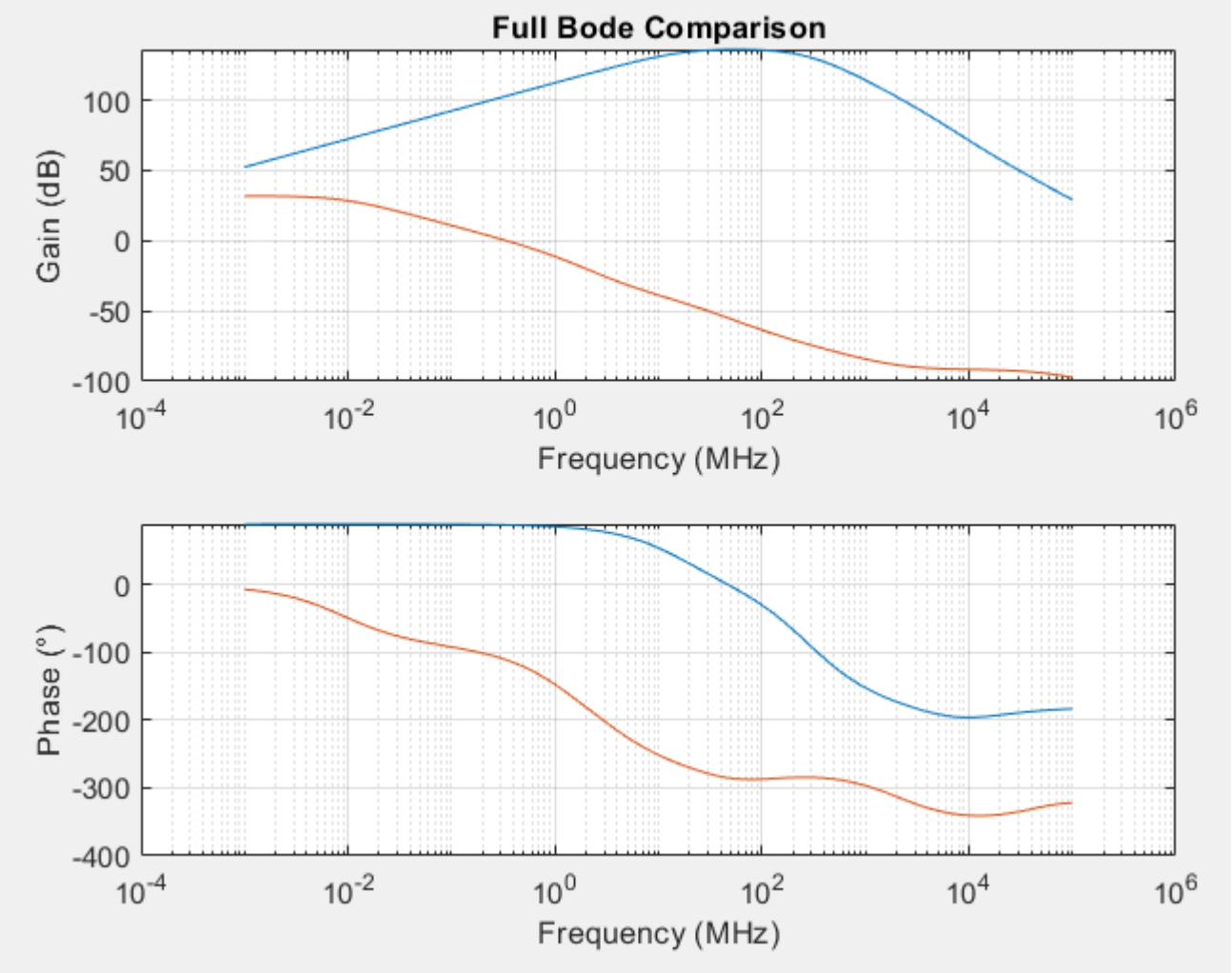
In the LTspice, the length of Mp8, Mp7 and Mp5 cannot set 1 nm with error (effective length <0nm), so I set the 50nm at least.

I think it may have 3 reasons:

- My pole and zero have problem
- I didn't consider the width of MOS.
- The resistance is negative.

The comparison from matlab:


Zoomed-in Bode Comparison
Gain (dB)
100
50
0
-50
-100
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
Frequency (MHz)
Phase (°)
100
0
-100
-200
-300
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
Frequency (MHz)



Simulate A_v in LTspice for the following cases:

a. No compensation network ($C_M = R_M = 0$).

b. With compensation capacitor but no compensation resistor ($R_M = 0$).

c. With both the compensation capacitor and the compensation resistor.

Then, show all 3 cases in seperate plots such that the differences are clearly visible. Furthermore, indicate on each plot the resulting PM.

Plot:

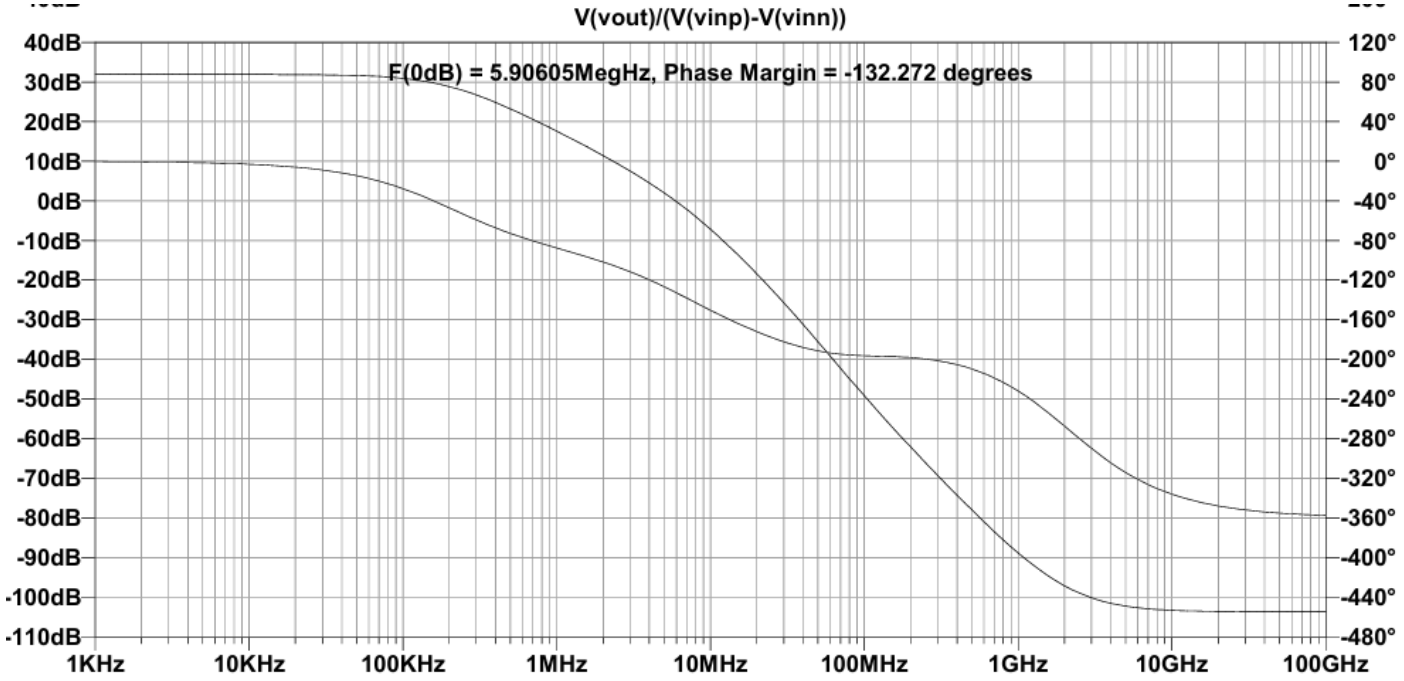
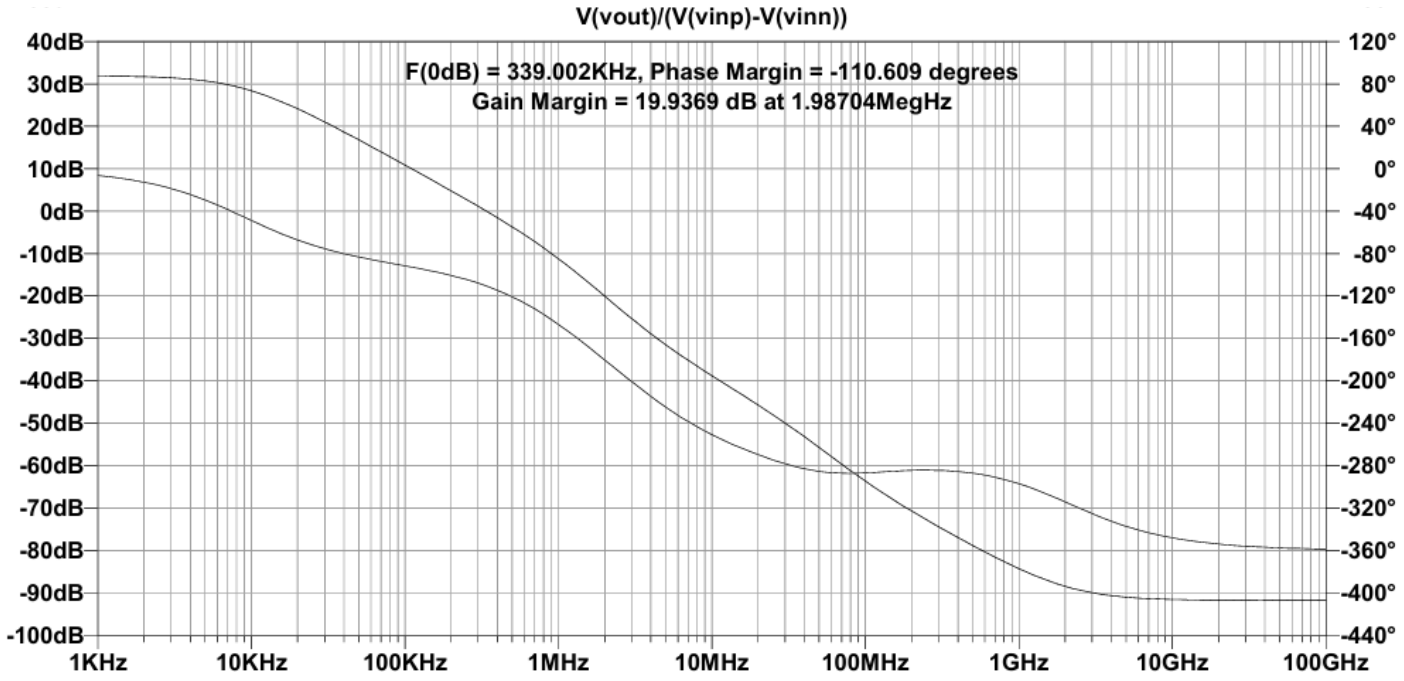
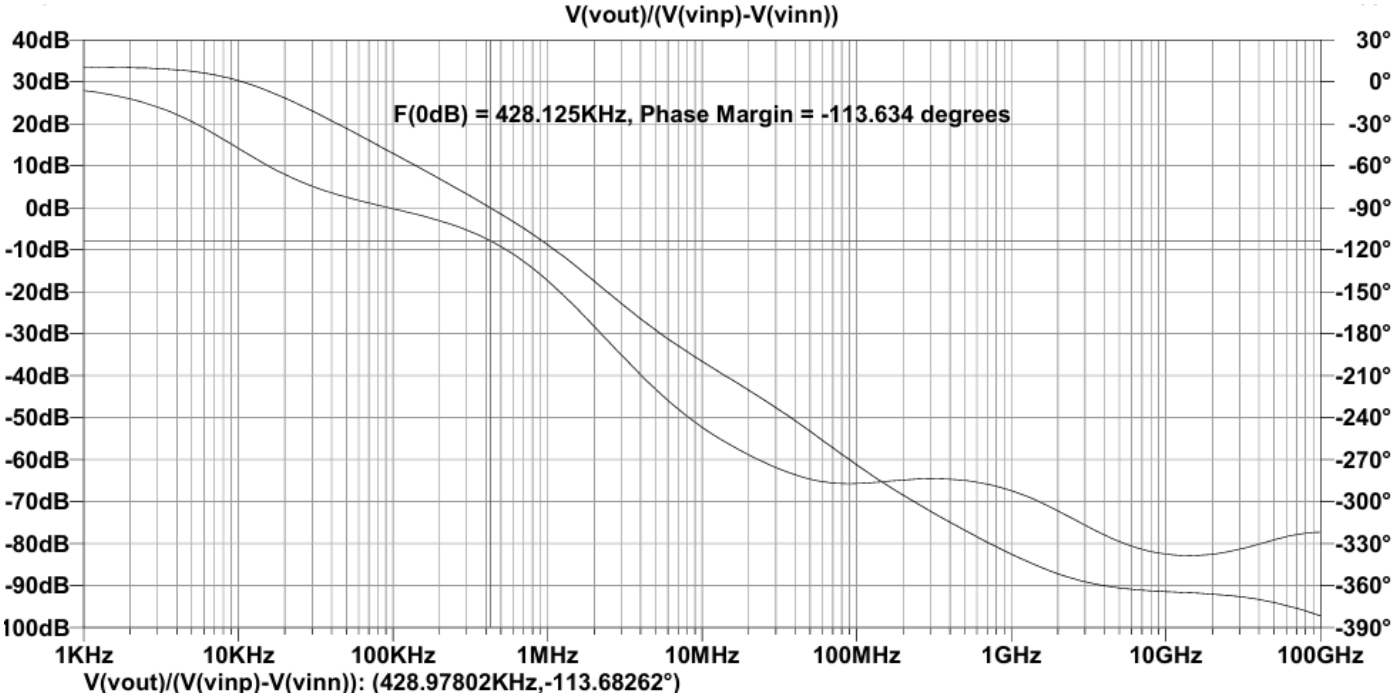


Figure : No compensation network



Figrue: No compensation resistor



Figrue: With both compensation components

Case	PM [deg]
No compensation network	47.73
No compensation resistor	69.4
With both compensation components	67

Explain, analyse and interpret the results above (for each case, explain the effect it has on the plots and PM.

B. Noise contribution in LTspice

Simulate the output-referred noise voltage power density $\overline{dv_{out,eq}^2}/\Delta f$ over an appropriate frequency range in LTspice. Then, insert the $\overline{dv_{out,eq}^2}/\Delta f$-plot below (use a logarithmic plot).

Plots:

Explain, analyse and interpret the results above (what regions can you see in the plot, and why?).

Simulate the input-referred noise spectral density by using the .NOISE option in LTspice. Also, by using LTspice, determine the total output integrated noise of the OpAmp. For that, take the integration bandwidth from 10 kHz to $10 \cdot f_{GBW}$. Furthermore, copy and paste the top 9 contributing elements using "View->SPICE Error Log" option and insert that list below.

Explain, analyse and interpret the results in above (what regions can you see in the plot, and why? + why is every transistor ranked the way it is for noise contribution).

C. Linearity

Explain why gain compression occurs when you increase the input voltage.

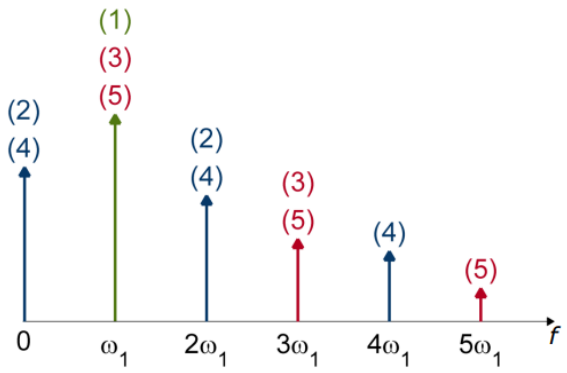
In a low-signal (small-signal) region, the amplifier behaves linearly: output voltage scales with input by a constant gain A. As Vin grows and drives the amplifier's output near its rails or transistor limits, additional increments in Vin yield smaller increments in Vout than the linear gain would predict.

1. Nonlinearities in the device: hormonic distortion.

One-tone analysis of a static nonlinearity

System: $y(t) = K_1 x(t) + K_2 x^2(t) + K_3 x^3(t) + \cdots$

Input: $x(t) = A \cdot \cos(\omega_1 t + \psi_0)$



Assume an input voltage is applied such that the full voltage swing (as calculated) is utilized. Does increasing the input voltage still increase the output voltage at this point? Explain.

Supply/Voltage Swing Limits: The output voltage of the amplifier can only be close to the power supply voltage at the highest and close to the ground or negative power supply at the lowest. If the input is larger, there is no extra "headroom" for the output voltage to continue to rise or fall, so the output has only a little or almost no margin for change, which eventually manifests as: the output signal becomes fixed and can no longer be expanded in a linear relationship.

BONUS: I didn't set up my large-signal simulation parameters correctly and my gain vs Vin plot is first rising, then stays flat and finally it starts dropping. What would I have expected to see and what could have gone wrong?

Conclusion (10 points):

Conclude your experiment by filling the editable fields in the performance table depicted below

Metric	Units	Specification	from hand-calculations	MATLAB	LTspice
DC gain	magnitude	223.87	223.87	13.8	85
DC gain	dB	47	47	4.9	38.6
Gain-Bandwidth frequency f_{GBW}	MHz	50	50	43.2	2.6/2pi=0.41
Dominant pole frequency f_{dom}	kHz	1727.88	1727.88	16554.7	2710
PM	°	> 70	80	1	67
$V_{in,cm,max} - V_{in,cm,min}$	V		0.622	0.310	0.89
$V_{out,max} - V_{out,min}$	V	> 0.7	0.72	0.774	
P_{diss}	mW			3.6	
$\overline{v_{out,eq}^2}$ (output-referred noise)	V_{rms}				

Comment about the deviations between hand calculation, MATLAB and LTspice values found above. Conclude, what the causes of such deviations are.

1. My pole and zero have problem
2. I didn't consider the width of MOS.
3. The resistance is negative.

If you have results which are not passing the specifications: conclude for each of those result what the cause is and what you need to do in order to make it pass, if you were repeating this experiment.

Issue1: I obfuscate the poles. In the specifications, can get pole 2 from PM and $|p2/GBWT$ vs Phase Margin in Two-poles System. And I also can get the pole 2 from GBW(the frequency in the 0 dB in the open loop and –3dB in the closed loop)= Av1*Av2* Dominant pole. Therefore, in the specifications, the Dominant pole frequency = 5.5(80degree)*314.16= 1727.88M rad/s; In the hand calculation, GBW= Av1*Av2*Dominant pole, then I get Dominant pole is 1.47M Rad/s. I think the result from $|p2|\ GBWT$ vs Phase Margin in Two-poles System is not my Dominant pole. That is p3 in the matlab.

Issue 2: I should set the different wigth in the Mp 5, Mp7 and Mp8. They have different mirror ratios causing the different widths in the Mp 5, Mp7 and Mp8. From the gm=2*ID/Vov, gm6=15.7mS, gm1=0.78mS. I can get the IDC,s1 and IDC,s2= 78uA , 57mA. then use the ratio= IDC,s1/Ibias=78uA/56.8uA=1.37, ratio= IDC,s1/Ibias=1.57mA/56.8uA. So the W/L of Mp7 is 1.37*W/L of Mp8, the W/L of Mp5 is 27.6*W/L of Mp8.

Issue 3: i set the stage 2 pole 3 is too high, p2=5.5（80 degree)*GBW, it's hard to set the MOS in the Ibias and keep the MOS in the saturation. I can set the p2=3.5(75degree)*GBW.

Issue 4: when I design the NMOS and PMOS, I should change the Vds, I cannot keep the Vds of Mp7, Mp2 and Mp4 equaling 0.55v at same times because I don't have enough VDD to support. and i should always keep the VDS ≥VGS −Vth =Vov in Nmos and |VSD |≥|VSG −Vth,p |=|Vov,p |in Pmos.

Issue5: when i calculate Vg5, i should get Vth,p from code =-0.22V. |VSG5|≈|Vth,p |+|Vov5 |≈0.22+0.2=0.42V
Vg5=Vdd-VSG=1.1-0.42=0.68V

New version:
P3=1727.88M
P2=1.47M (Dominant pole)

However i still cannot right Rm

```
p2 = 1.47e6; % p2 = 1.47 * 10^6
Mn6_gm = 15.7e-3; % Mn6.gm = 15.7 * 10^-3
cm = 2.5e-12; % cm = 2.5 * 10^-12
%spec.Rm
spec_Rm = (1 / Mn6_gm) - (1 / (cm * p2));
% show reuslt
fprintf('spec.Rm = %.4f Ohms\n', spec_Rm);
```

%W=1/(Cm*(1/gm6-Rm) to get Rm= -272045 Ohms

hand calculation:

$C_m = 2.5pF = 2.5\times10^{-12}F$ DC gain $= 47dB = 10^{\frac{47}{20}} = 223.87$ mag.

$C_L = 10pF = 10\times10^{-12}F$ $f_{GBW} = 50MHz = 50\times10^6 Hz$

$PM \geq 70 = 80$ degree. Output swing $> 0.7V$

from: $|P_2|/GBW_T$ Vs Phase Margin graph: 80 degree $\Rightarrow \frac{|P_2|}{GBW} = 5.5$

DC gain: $A_V = A_{V1}\times A_{V2} = 223.87$ mag.

$A_{V1} = -\frac{g_{m1,2}}{g_{ds4}+g_{ds2}} = \frac{V_{in}\cdot -g_{m1,2}}{g_{ds2}+g_{ds4}} \Rightarrow GA_{V1} = -\frac{g_{m1,2}}{g_{ds4}+g_{ds2}}$ I set $g_{ds4}=g_{ds2}$

$\therefore A_{V1} = -\frac{g_{m1,2}}{2g_{ds2}}$

for stage 1: $V_{stage1\,out} = I\cdot R_{out_1} = \frac{I}{g_{out1}}$

node: I_2 ↑, V_{out}, I_4 ↓

A_{V2}: V_{in2} ↑, I_5, V_{out2}, I_6 ↑

Same as A_{V1}: $\therefore G_{AV2} = \frac{V_{out}}{V_{in}} = \frac{-g_{m6}}{g_{ds5}+g_{ds6}}$

I set $g_{ds5}=0$, $\therefore A_{V2} = -\frac{g_{m6}}{g_{ds6}}$

$P_2 = \frac{g_{ds2}\parallel g_{ds4}}{C_M(1-V_{A2})} = \frac{\frac{1}{2}g_{ds2}}{C_M(1-V_{A2})} = \frac{\frac{1}{2}g_{ds2}}{C_m\cdot\frac{g_{m6}}{g_{ds6}}}$

biggest, ignore 1, $V_{A2} >> 1$,

$GBW = 2\pi\cdot f_{GBW} = 2\cdot pi\times50M = 314.16$ Mrad/s $= 10^{-}$

$GBW = A_{V1}\cdot A_{V2}\cdot P_2 = -\frac{g_{m1,2}}{2g_{ds2}}\cdot\frac{-g_{m6}}{g_{ds6}}\cdot\frac{\frac{1}{2}\cdot g_{ds2}}{C_m\cdot\frac{g_{m6}}{g_{ds6}}} = \frac{g_{m1,2}}{C_M} = 314.16$ Mrad/s

$g_{m1,2} = 314.16\times10^6\cdot2.5\times10^{-12} = 0.000785 = 7.85$ ms

$P_3 = \frac{g_{m6}}{C_L} = 5.5\times GBW = 1728$ Mrad/s $\Rightarrow g_{m6} = 1728\times10^6\cdot10\times10^{-12} = 0.0173$

$= 17.3$ ms

As the $C_{P1,P2} << C_L$, or C_m, $W_{P1,P2}$ is the fast pole, cannot be dominant,

I set nodes before. C_m be the dominant pole, P_3 is before V_{out}, C_{load}.

P_2

```
%% Analog Electronics final project
clc;
close all;
clear;
addpath(genpath('circuitDesign'));
addpath(genpath('models'));
load('UMC65_RVT.mat');
%% Initializations
designkitName       = 'umc65';
circuitTitle        = 'Analog Design - Project';
elementList.nmos    = {'Mn3','Mn4','Mn6'};
elementList.pmos    = {'Mp1','Mp2','Mp5','Mp7','Mp8'};
choice.maxFingerWidth = 10e-6;
choice.minFingerWidth = 200e-9;
simulator                   = 'spectre';
simulFile                   = 0;
simulSkelFile       = 0;
spec                        = [];
analog              = cirInit('analog',circuitTitle,'top',elementList,spec,choice,...
designkitName,NRVT,PRVT,simulator,simulFile,simulSkelFile);
analog              = cirCheckInChoice(analog, choice);
%% Project: circuit
disp(' ');
disp(' VDD VDD VDD ');
disp(' | | | ');
disp(' Mp8-+---------Mp7---------------------Mp5 ');
disp(' |--+ | | ');
disp(' | +--+--+ node 3-> +-----+---OUT ');
disp(' | | | | | ');
disp(' | IN1--Mp1 Mp2--IN2 | | ');
disp(' | | | | | ');
disp(' | node 1-> | | <-node 2 | Cl ');
disp(' | |--+ +------+-Cm---Rm----+ | ');
disp(' Ibias | | | | ↑ | | ');
disp(' | Mn3-+-Mn4 | node 4 | | ');
disp(' | | | +-----------Mn6 | ');
disp(' | | | | | ');
disp(' GND GND GND GND GND ');
%% AI: Implement your OpAmp according to your handcalculations
Ibias=56.8e-6;
fGBW=55e6;
Cl=10e-12;
Cm=2.5e-12;% To have a good stability increasing Cc will increase stability
%(page 15 chaptre 8)
GBW=fGBW*2*pi;
VDD=1.1;
%%Mp1
Mp1.lg=100e-9;
```

```
Mp1.vov=-0.43;
Mp1.gm=GBW*Cm;
Mp1.vds=-0.2;
Mp1.vsb=0;
Mp1.vth=tableValueWref('vth',PRVT,Mp1.lg,0,Mp1.vds,Mp1.vsb);
Mp1.vgs=Mp1.vth + Mp1.vov;
Mp1.w=mosWidth('gm',Mp1.gm,Mp1);
Mp1=mosNfingers(Mp1);
Mp1=mosOpValues(Mp1);
%%Mp2
Mp2.lg=Mp1.lg;
Mp2.vov=Mp1.vov;
Mp2.gm=Mp1.gm;
Mp2.vds= Mp1.vds;
Mp2.vsb=0;
Mp2.vth=tableValueWref('vth',PRVT,Mp2.lg,0,Mp2.vds,Mp2.vsb);
Mp2.vgs=Mp2.vth + Mp2.vov;
Mp2.w=mosWidth('gm',Mp2.gm,Mp2);
Mp2=mosNfingers(Mp2);
Mp2=mosOpValues(Mp2);
%%Mn6
Mn6.lg=80e-9;
Mn6.vov=0.18;
%Mn6.vov=0.2;
Mn6.gm=GBW*Cl/tan(deg2rad(10));
Mn6.vds=0.55;
Mn6.vsb=0;
Mn6.vth=tableValueWref('vth',NRVT,Mn6.lg,0,Mn6.vds,Mn6.vsb);
Mn6.vgs=Mn6.vth + Mn6.vov;
Mn6.w=mosWidth('gm',Mn6.gm,Mn6);
Mn6=mosNfingers(Mn6);
Mn6=mosOpValues(Mn6);
%%Mn3
Mn3.lg=500e-9;
Mn3.ids=Mp1.ids;
Mn3.vds=Mn6.vgs;
Mn3.vsb=0;
Mn3.vgs=Mn6.vgs;
Mn3.vth=tableValueWref('vth',NRVT,Mn3.lg,0,Mn3.vds,Mn3.vsb);
%Mn3.vgs=Mn3.vth + Mn3.vov;
Mn3.vov=Mn3.vgs+Mn3.vth;%
Mn3.w=mosWidth('ids',Mn3.ids,Mn3);
Mn3=mosNfingers(Mn3);
Mn3=mosOpValues(Mn3);
%%Mn4
Mn4.lg=Mn3.lg;
Mn4.ids=Mn3.ids;%microAmper
Mn4.vds=Mn3.vds;
Mn4.vsb=0;
Mn4.vgs=Mn6.vgs;
Mn4.vth=tableValueWref('vth',NRVT,Mn4.lg,0,Mn4.vds,Mn4.vsb);
%Mn4.vgs=Mn4.vth + Mn4.vov;
Mn4.vov=Mn4.vgs-Mn4.vth;
```

```
Mn4.w=mosWidth('ids',Mn4.ids,Mn4);
Mn4=mosNfingers(Mn4);
Mn4=mosOpValues(Mn4);
%%Mp5
Mp5.lg=1000e-9;
Mp5.vov=-0.2;
Mp5.ids=Mn6.ids;%microAmper
Mp5.vds=-0.2;
Mp5.vsb=0;
Mp5.vth=tableValueWref('vth',PRVT,Mp5.lg,0,Mp5.vds,Mp5.vsb);
Mp5.vgs=Mp5.vth + Mp5.vov;
Mp5.w=mosWidth('ids',Mp5.ids,Mp5);
Mp5=mosNfingers(Mp5);
Mp5=mosOpValues(Mp5);
%%Mp7
Mp7.lg=1000e-9;
Mp7.vov=Mp5.vov;
Mp7.ids=Mp1.ids+Mp2.ids;%microAmper
Mp7.vds=-VDD-Mp1.vds+Mn3.vds;
Mp7.vsb=0;
Mp7.vth=tableValueWref('vth',PRVT,Mp7.lg,0,Mp7.vds,Mp7.vsb);
Mp7.vgs=Mp7.vth + Mp7.vov;
Mp7.w=mosWidth('ids',Mp7.ids,Mp7);
Mp7=mosNfingers(Mp7);
Mp7=mosOpValues(Mp7);
%%Mp8
Mp8.lg=1000e-9;
Mp8.vov=Mp5.vov;%weak inversion
%Mp8.gm=;
% Mp8.gds=20.26e-6;%microS
Mp8.ids=Ibias;%microAmper
Mp8.vds=-0.55;
Mp8.vsb=0;
Mp8.vth=tableValueWref('vth',PRVT,Mp8.lg,0,Mp8.vds,Mp8.vsb);
Mp8.vgs=Mp8.vth + Mp8.vov;
Mp8.w=mosWidth('ids',Mp8.ids,Mp8);
Mp8=mosNfingers(Mp8);
Mp8=mosOpValues(Mp8);
%% AI: Fill out the empty variables required to plot the transfer-function.
% meaning of each variable see comment and
% location of nodes see line 31
AvDC1 =(Mp2.gm)/(Mp2.gds+Mn4.gds);
AvDC2 =(Mn6.gm)/(Mp5.gds+Mn6.gds); % DC gain 2nd stage
C1 =Mn3.cdb+Mn3.cgb+Mn3.cgs+Mp1.cdb+Mp1.cgd+Mn4.cgd+Mn4.cgb+Mn4.cgb ; % Capacitance on node 1
G1 = Mn3.gm; % Admittance on node 1
C2 =Cm*(1+AvDC2)+Mn4.cdb+Mn4.cgd+Mp2.cgd+Mp2.cdb+Mn6.cgd+Mn6.cgs+Mn6.cgb; % Capacitance on node
2
G2 = Mp2.gds+Mn4.gds; % Admittance on node 2
C3 = Cl; % Capacitance on node 3
G3 = Mn6.gm; % Admittance on node 3
C4 = Cm*(1+AvDC2)+Mn6.cdb+Mn6.cgb+Mn6.cgs; % Capacitance on node 4
G4 = Mn6.gm; % Admittance on node 4 (hint: what happens with CL at very high frequencies?)
%% AI: Set-up Rm, Cc and CL and calculate the zero required for the transfer-fct
```

```
spec.Cm = 2.5e-12;
spec.Cl = 10e-12;
spec.Rm = -173.7;
z2 = 1/(1/spec.Rm-Mn6.gm); % expression of zero due to the nulling resistor [rad/s]
%% AI: Fill out the empty variables required for the performance summary
Vin_cm_min=50e-3+Mp1.vgs+Mn3.vgs;
Vin_cm_max=VDD+Mp1.vgs+Mp7.vov;
Vout_cm_min = Mn6.vdsat;
Vout_cm_max = VDD - abs(Mp5.vdsat);
Pdiss = VDD * (Mp5.ids + Mp7.ids + Mp8.ids);
%% Sanity check (do not modify)
disp('=====================================');
disp('= Transistors in saturation =');
disp('=====================================');
if mosCheckSaturation(Mp1)
fprintf('\nMp1:Success\n')
end
if mosCheckSaturation(Mp2)
fprintf('Mp2:Success\n')
end
if mosCheckSaturation(Mn3)
fprintf('Mn3:Success\n')
end
if mosCheckSaturation(Mn4)
fprintf('Mn4:Success\n')
end
if mosCheckSaturation(Mp5)
fprintf('Mp5:Success\n')
end
if mosCheckSaturation(Mn6)
fprintf('Mn6:Success\n')
end
if mosCheckSaturation(Mp7)
fprintf('Mp7:Success\n')
end
if mosCheckSaturation(Mp8)
fprintf('Mp8:Success\n\n')
end
%% Summary of sizes and biasing points (do not modify)
disp('=====================================');
disp('= Sizes and operating points =');
disp('=====================================');
analog = cirElementsCheckOut(analog); % Update circuit file with
% transistor sizes
mosPrintSizesAndOpInfo(1,analog); % Print the sizes of the
% transistors in the circuit file
fprintf('IBIAS\t= %6.2fmA\nRm\t= %6.2f Ohm\nCm\t= %6.2fpF\n\n',Mp8.ids/1e-3,spec.Rm,spec.Cm/1e-
12);
%% Performance summary (do not modify)
disp('=====================================');
disp('= Performance =');
disp('=====================================');
fprintf('\nmetric \t result\n');
```

```
fprintf('Vin,cm,min [mV] \t%.0f\n',Vin_cm_min/1e-3);
fprintf('Vin,cm,max [mV] \t%.0f\n',Vin_cm_max/1e-3);
fprintf('Vout,cm,min [mV] \t%.0f\n',Vout_cm_min/1e-3);
fprintf('Vout,cm,max [mV] \t%.0f\n',Vout_cm_max/1e-3);
fprintf('Pdiss [mW] \t%.1f\n\n',Pdiss/1e-3);
disp('====================================');
disp('= Poles/Zeros =');
disp('====================================');
fprintf('\np1 = %.1f\tMHz\n', G1/(2*pi*C1*1e6));
fprintf('p2 = %.1f\tkHz\n', G2/(2*pi*C2*1e3));
fprintf('p3 = %.1f\tMHz\n', G3/(2*pi*C3*1e6));
fprintf('p4 = %.1f\tGHz\n', G4/(2*pi*C4*1e9));
fprintf('z1 = %.1f\tMHz\n', 2*G1/(2*pi*C1*1e6));
fprintf('z2 = %.1f\tMHz\n', z2);
%% Ploting transfer function (do not modify)
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% if control toolbox in Matlab is available
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
s = tf('s');
% transfer function
TF1 = AvDC1*AvDC2*((1+s*C1/(2*G1))*(1-s*(1/z2)))/ ...
((1+s*C1/G1)*(1+s*C2/G2)*(1+s*C3/G3)*(1+s*C4/G4));
freq = logspace(1,12,1e3);
figure(1)
bode(TF1,2*pi*freq); grid on;
h = gcr;
setoptions(h,'FreqUnits','Hz');
title('Frequency response Opamp');
hold all
spice_tbl = read_LTspice_data("final_project_data.txt");
[matlab_mag, matlab_phase] = bode(TF1, 2*pi*spice_tbl.freq);
figure
subplot(2, 1, 1);
semilogx(spice_tbl.freq*1e-6, 20*log10(squeeze(matlab_mag)));
hold on;
semilogx(spice_tbl.freq*1e-6, 20*log10(abs(spice_tbl.G1)));
grid on;
xlabel('Frequency (MHz)')
ylabel('Gain (dB)')
title('Full Bode Comparison')
subplot(2, 1, 2);
semilogx(spice_tbl.freq*1e-6, squeeze(matlab_phase));
hold on;
semilogx(spice_tbl.freq*1e-6, unwrap(angle(tbl.G1))*180/pi);
grid on;
xlabel('Frequency (MHz)')
ylabel('Phase (°)')
fGBW = Mp2.gm / (2*pi*spec.Cm);
[~, minidx] = min(abs(2*fGBW - spice_tbl.freq));
roi = 1:(minidx+1);
figure
subplot(2, 1, 1);
semilogx(spice_tbl.freq(roi)*1e-6, 20*log10(squeeze(matlab_mag(:, :, roi))));
```

```
hold on;
semilogx(spice_tbl.freq(roi)*1e-6, 20*log10(abs(spice_tbl.G1(roi))));
grid on;
xlabel('Frequency (MHz)')
ylabel('Gain (dB)')
title('Zoomed-in Bode Comparison')
subplot(2, 1, 2);
semilogx(spice_tbl.freq(roi)*1e-6, squeeze(matlab_phase(:, :, roi)));
hold on;
semilogx(spice_tbl.freq(roi)*1e-6, unwrap(angle(tbl.G1(roi)))*180/pi);
grid on;
xlabel('Frequency (MHz)')
ylabel('Phase (°)')
```

Appendix:

Previously, you were expected to include your MATLAB code and LTspice netlist here, but now we changed it so that you must upload them in the Canvas assignment. Read the description of the assignment for details on exactly what files are needed!

- D:\VUB\second year\analog electric\Matlab\LTspice\final_project.asc vdd N001 0 1.1 vcm N006 0 0.989 E1 vinn N006 N006 vinp 1 V1 vinp N006 SINE(0 100u 1k) AC 10m Mp8 N002 N002 N001 N001 p_11_sprvt l=50n w=18.46u Mp7 N003 N002 N001 N001 p_11_sprvt l=50n w=215.48u Mp1 N005 vinn N003 N003 p_11_sprvt l=100n w=4.47u Mp2 vinter vinp N003 N003 p_11_sprvt l=100n w=4.47u Mn4 vinter N005 0 0 n_11_sprvt l=500n w=6.33u Mn3 N005 N005 0 0 n_11_sprvt l=500n w=6.33u Mn6 vout vinter 0 0 n_11_sprvt l=80n w=25.3u Mp5 vout N002 N001 N001 p_11_sprvt l=50n w=1.02u Rm vout N004 -173 Cm N004 vinter 2.5p Ibias N002 0 0.06m Cload vout 0 10p .model NMOS NMOS .model PMOS PMOS .lib C:\Users\wangw\AppData\Local\LTspice\lib\cmp\standard.mos .include BSIM4_UMC65.lib
- .op .ac dec 20 1k 100G .backanno .end